

Microsaccades and temporal attention at different locations of the visual field

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Temporal attention, the prioritization of information at specific points in time, improves performance in behavioral tasks but cannot ameliorate the perceptual asymmetries that exist across the visual field. That is, even after attentional deployment, performance is better along the horizontal than vertical meridian and worse at the upper than lower vertical meridian. Here we asked whether and how microsaccades—tiny fixational eye-movements—could mirror or alternatively attempt to compensate for these performance asymmetries by assessing temporal profiles and direction of microsaccades as a function of visual field location. Observers were asked to report the orientation of one of two targets presented at different time points, in one of three blocked locations (fovea, right horizontal meridian, upper vertical meridian). We found the following: (1) Microsaccade occurrence did not affect either task performance or the magnitude of the temporal attention effect. (2) Temporal attention modulated the microsaccade temporal profiles, and this modulation varied with polar angle location. At all locations, microsaccade rates were significantly more suppressed in anticipation of the target when temporally cued than in the neutral condition. Moreover, microsaccade rates were more suppressed during target presentation in the fovea than in the right horizontal meridian. (3) Across locations and attention conditions, there was a pronounced bias toward the upper hemifield. Overall, these results reveal that temporal attention benefits performance similarly around the visual field, microsaccade suppression is more pronounced for attention than expectation (neutral trials) across locations, and the directional bias toward the upper hemifield could reflect an attempt to

compensate for typical poor performance at the upper vertical meridian.

Introduction

Visual attention, be it spatial, temporal or feature-based, enables us to selectively process information in our environment—at particular locations or points in time or for specific features—to optimize behavior. Research in the temporal domain has been often clouded by the synonymous use of “attention” and “expectation.” But they are dissociable processes elicited by different task demands (Wyart, Nobre, & Summerfield, 2012; Summerfield & de Lange, 2014; Bang & Rahnev, 2017; Rungratsameetaweemana & Serences, 2019). Temporal attention allows us to highlight behaviorally-relevant information at a specific point in time to improve performance (Nobre, Correa, & Coull, 2007; Nobre & Rohenkohl, 2014; Denison, Heeger, & Carrasco, 2017; Nobre & van Ede, 2018; Denison, Yuval-Greenberg, & Carrasco, 2019; Denison, Carrasco, & Heeger, 2021), whereas temporal expectation is the ability to recognize stimulus timing patterns regardless of behavioral relevance (Rohenkohl, Cravo, Wyart, & Nobre, 2012; Denison et al., 2019). Similar to the benefits and costs seen for endogenous (voluntary) and exogenous (involuntary) spatial attention (e.g., Pestilli & Carrasco, 2005; Montagna, Pestilli, & Carrasco, 2009; Fernández, Okun, & Carrasco, 2022), endogenous temporal attention enhances target discriminability at a cued point in time and impairs performance at earlier and later time points

Citation: Palmieri, H., Fernández, A., & Carrasco, M. (2023). Microsaccades and temporal attention at different locations of the visual field. *Journal of Vision*, 23(5):6, 1–17, <https://doi.org/10.1167/jov.23.5.6>.



(Denison et al., 2017; Fernández, Denison, & Carrasco, 2019; Denison et al., 2021).

Performance varies across (reviews, Anton-Erxleben & Carrasco, 2013; Strasburger, Rentschler, & Jüttner, 2011) and around the visual field (review, Himmelberg, Winawer, & Carrasco, 2023). Performance fields refer to the behavioral asymmetries at isoeccentric locations around the visual field (e.g., Carrasco, Talgar, & Cameron, 2001; Talgar & Carrasco, 2002; Baldwin, Meese, Baker, 2012; Barbot, Xue, & Carrasco, 2021). One asymmetry is the horizontal-vertical anisotropy, in which performance is superior along the horizontal than the vertical meridian. The other is vertical meridian asymmetry, in which performance is better on the lower than the upper vertical meridian. These differences are also present with regard to information accrual—information is processed more quickly along the horizontal than the vertical meridian and along the lower than the upper vertical meridian (Carrasco, Giordano, & McElree, 2004). The discriminability effects are widespread and are present when the target is presented alone or amid distractors (Purokayastha, Roberts, & Carrasco, 2021), under monocular or binocular conditions (Carrasco et al., 2001; Barbot et al., 2021) and on retinotopic coordinates regardless of head rotation (Corbett & Carrasco, 2011). These asymmetries increase with eccentricity and spatial frequency (e.g., Carrasco et al., 2001; Baldwin, Meese, Baker, 2012; Himmelberg, Winawer, & Carrasco, 2020).

These polar angle asymmetries are resilient to the effects of covert attention: Both exogenous (Carrasco et al., 2001; Carrasco, Williams, & Yeshurun, 2002; Roberts, Ashinoff, Castellanos, & Carrasco, 2018) and endogenous (Purokayastha et al., 2021) spatial attention similarly improves performance around the visual field. Comparable effects have been reported in the temporal domain, a study investigating the effects of temporal attention revealed that temporal attention also improves behavioral performance (d') and reduces reaction times to a similar extent across a subset of performance field locations—fovea, right-horizontal meridian (RHM), and upper-vertical meridian (UVM)—thus maintaining the performance field asymmetries (Fernández et al., 2019). Here we investigated fixational eye movements in that study to further characterize how temporal attention and performance fields interact with one another.

Microsaccades are small ($<1^\circ$ of visual angle), involuntary eye movements made during fixation ($\sim 2/s$) (Martínez-Conde, Macknik, & Hubel, 2004; Collewyn & Kowler, 2008; review, Rolfs, 2009; Rucci & Poletti, 2015). Whereas a discrimination response provides information only at the end of the trial, microsaccades provide an analogous online window measure into cognitive and perceptual processes throughout the trial. In addition to preventing perceptual fading (e.g., Martínez-Conde, Macknik, Troncoso, & Dyar,

2006), they play a role in many tasks, for example in high-acuity vision (Rucci, Iovin, Poletti, & Santini, 2007; Ko, Poletti, & Rucci, 2010) and perceptual learning (Hung & Carrasco, 2022), and have been assessed in neurotypical and special populations (e.g., people with attention deficit hyperactivity disorder [ADHD]; Fried et al., 2014; Dankner, Shalev, Carrasco, & Yuval-Greenberg, 2017; Roberts et al., 2018).

The role of microsaccades in spatial covert attention tasks has been debated; some assert they play a critical role, that they are linked to shifts in spatial attention (Yuval-Greenberg, Merriam, & Heeger, 2014; Hafed, Chen, & Tian, 2015). Moreover, monkey V4 and IT cortices show that firing rates and stimulus representations were enhanced by attention when followed by a microsaccade directed toward the attended target (Lowet et al., 2018). Whereas others have questioned the role of microsaccades and posit that they may play a role but are not necessary, because attention-related modulation in the superior colliculus occurs even in the absence of microsaccades (Yu, Herman, Katz, & Krauzlis, 2022) and the lack of attention-driven microsaccades does not inhibit alpha activity modulation (Liu, Nobre, & van Ede, 2022). Furthermore, others have shown covert attention effects in the absence of any microsaccades (Poletti, Rucci, & Carrasco, 2017) or reported similar attention effects in the presence or absence of microsaccades (Roberts & Carrasco, 2019; Li, Pan, & Carrasco, 2021).

Only one study has investigated the role of microsaccades in temporal attention. Deploying temporal attention at the time of a brief validly-cued visual stimulus suppresses microsaccades beyond the suppression brought about by the effects of temporal expectation (Denison et al., 2019). Microsaccades occurring during a brief stimulus presentation would hinder performance. Indeed, in anticipation of a visual target onset (i.e., temporal expectation), microsaccade rate is suppressed to avoid retinal slippage, which would cause transient blur or backward masking (Dankner et al., 2017; Amit, Abeles, Carrasco, & Yuval-Greenberg, 2019). This microsaccade suppression is also found during anticipation of auditory (Abeles, Amit, Tai-Perry, Carrasco, & Yuval-Greenberg, 2020) and tactile (Badde, Myers, Yuval-Greenberg, & Carrasco, 2020) stimuli, indicating that microsaccadic inhibition is a marker of supramodal temporal expectation. Together these results suggest that microsaccades hinder perceptual processing in different modalities while the stimulus is briefly presented and thus are minimized by the oculomotor system.

In contrast, microsaccades are beneficial in some tasks. For example, given that vision is not uniform within the foveola, microsaccades compensate for this lack of homogeneity. For high-acuity tasks with stimuli presented for one second or more, precise

microsaccades relocations of gaze equate performance at different foveal eccentricities (e.g., Rucci, Iovin, Poletti, & Santini, 2007; Ko, Poletti, & Rucci, 2010; Poletti, Listeri, & Rucci, 2013). Microsaccades can also be directed to a memorized location of a stimulus during memory selection, suggesting that microsaccades can assist in internal information readout (van Ede, Chekroud, & Nobre, 2019; Hung & Carrasco, 2022).

Here we investigated whether and how the dynamics and direction of microsaccades vary under temporal attention manipulations at selected polar angle locations of the visual field (fovea and parafoveal RHM and UVM) where discriminability and speed of information accrual vary substantially under regular viewing conditions. We hypothesized that microsaccade rates would be more reduced during the critical period (within 100 ms before target onset; as in Denison et al., 2019) for attention than neutral conditions. Furthermore, we assessed whether microsaccade direction would be biased toward target location, and we evaluated whether microsaccades could mirror performance asymmetries (i.e., be more frequent) toward locations with high performance (e.g., along the horizontal meridian), or alternatively could attempt to play a “compensatory” role (i.e., be more frequent) toward locations with typical poor performance (e.g., at the upper vertical meridian).

Method

Observers

The observers were the same 10 observers (five female, aged 21–33) reported in the analysis of Fernández et al. (2019). The eye data for one participant could not be included because of problems with data recovery during the microsaccade extraction pipeline. All observers had normal or corrected vision and provided written consent before testing. All experimental procedures were in line with the Helsinki declaration, and the University Committee on Activities Involving Human Subjects at New York University approved all experimental protocols.

Stimulus

Stimuli were generated on an Apple iMac computer using MATLAB (MathWorks, Natick, MA, USA) and the PsychToolBox (Brainard, 1997; Pelli, 1997). Stimuli were displayed on a gamma-calibrated CRT monitor (1280 × 960 resolution; 100 Hz refresh rate) 57 cm away from the observer in a dark room. Observers’ heads were stabilized by a chin rest. Fixation was a white “X” subtending 0.5° of visual angle. Stimuli were four cycles

per degree sinusoidal gratings with a Gaussian spatial envelope ($SD = 0.7^\circ$). Stimuli were 100% contrast, presented on a medium-gray background (57 cm^2), placed in one of three locations in a blocked design to eliminate spatial uncertainty: at fixation (FOVEA), at 4° along the RHM, or at 4° along the UVM. Two targets (temporal interval 1 and 2 [T1 and T2]) were presented in each trial, and each target was independently tilted either counterclockwise or clockwise with respect to either horizontal or vertical. Stimulus placeholders were the corners of a $4.25^\circ \times 4.25^\circ$ white square outline with a linewidth of 0.08° centered on the target location. These placeholders were present throughout the trial and during all trials to eliminate spatial uncertainty. All the auditory cues (three possible tones: T1, T2, and neutral) consisted of pure sine wave tones played through the computer’s speakers.

Procedure

Observers were asked to discriminate the orientation of one of two Gabor patches (Figure 1). Each observer completed two sessions of each spatial location condition, resulting in six sessions per observer. Location sessions were performed in a random order. Each session consisted of five 64-trial blocks for a total of 320 trials, 1920 trials per participant. Trial order per block was counterbalanced to result in an equal number of trials per precue, target, and orientation.

At the start of each trial, an auditory precue was presented 1000 ms before the first target. It was either a valid cue, indicating which target (T1: 1300 Hz or T2: 250 Hz) to attend to, or an uninformative, neutral auditory precue (T1 and T2) with the two tones together. Two targets were presented serially for 30 ms at the same target location, separated by a stimulus onset asynchrony of 250 ms. A response-cue, 500 ms after the last stimulus presentation, informed the observer of which target orientation to report. For valid trials, the response-cue indicated the same target that the precue signaled, and on neutral trials, the response-cue indicated either T1 (high tone: 1300 Hz) or T2 (low tone: 250 Hz), with equal probability. Observers reported the tilt of the response-cued target with a key press (1: counter-clockwise, 2: clockwise). In neutral trials, the temporal precue indicated that the target was equally likely to be in the first or second interval and the response-cue indicated the interval containing the target whose orientation the participant had to report. To reduce speed-accuracy tradeoffs, observers were asked not to respond until the go-cue (1500 ms after response-cue offset) would be displayed, in which the fixation cross changed from white to gray. The response time period itself was unlimited. At the end of each trial, after each response, observers received feedback: a green plus sign for correct and

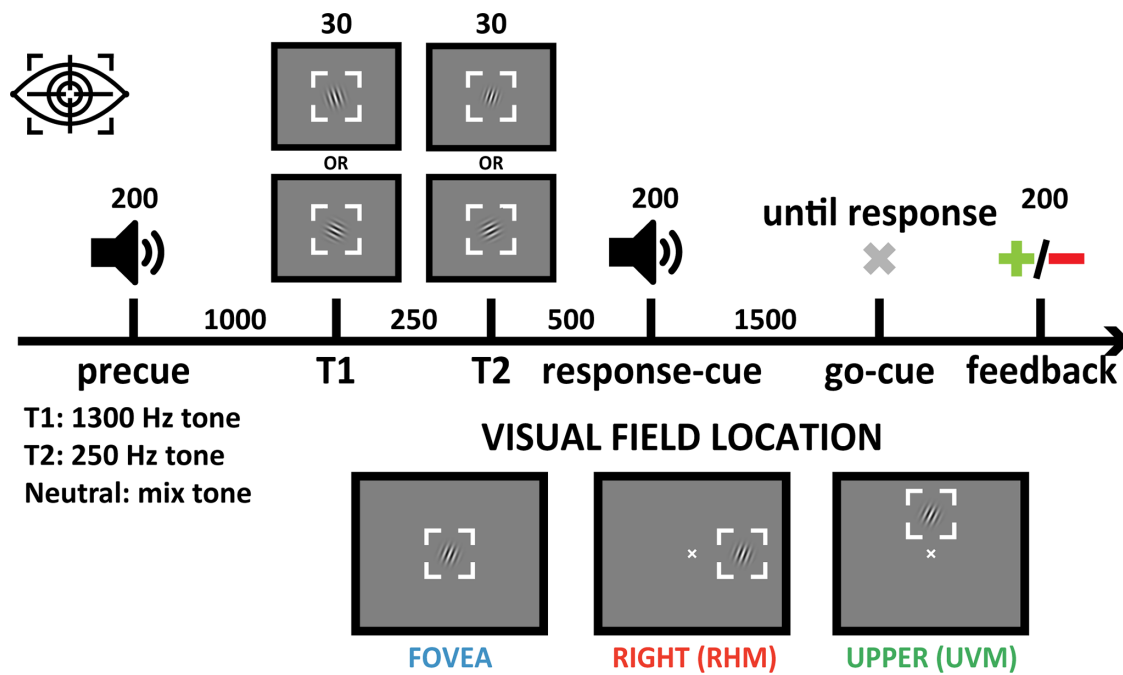


Figure 1. Example trial. Example of a foveal location trial sequence in milliseconds. Visualization of all location trial types. Observers performed a 2-Interval Forced Choice (2IFC) orientation discrimination task, while directed with a precue tone to deploy temporal attention to one of two (or both, neutral condition) Gabor patches presented serially (stimulus onset asynchrony: 250 ms) at the same blocked target location: fovea, RHM, or UVM at 4°.

red minus sign for incorrect responses. Block-level feedback was also provided, with percent correct for that block, to inform participants of their overall performance. Additionally, block-level performance was informative to the experimenter to confirm that the titrated threshold (and adjusted threshold, thereafter) would yield ~75% performance.

To accurately compare attentional effects among visual field locations, discriminability was matched across the three tested locations for each observer. Before each experimental session, an adaptive staircase (three-up/one-down thresholding staircase) procedure was used to titrate discrimination accuracy to 75% on neutral trials, independently per target (T1 and T2) and location (fovea, RHM, or UVM) resulting in six thresholds (80 trials per threshold). Each session began with the titrated threshold, and the threshold was adjusted per block as needed to ensure the same accuracy across the experiment. To familiarize observers with the task and target timing, each observer performed two training sessions of one hour at an experimentally irrelevant location (lower right quadrant) on neutral trials.

Eye tracking

To ensure that observers were centrally fixating and to analyze microsaccades, online eye tracking was used

at a sampling rate of 1000 Hz. Eye movements were monitored using an EyeLink 1000 Plus Desktop Mount eyetracker (SR Research, Ontario, Canada). Each trial began once the observer fixated on the central cross for 300 ms. If observers broke fixation (more than 1.5° from center) between the precue and the response-cue the trial would end and be added to the end of the block. Observers were allowed to move their eyes, blink, to rest their eyes between trials as needed.

Data analysis

We analyzed 15 sessions from the fovea ($n = 9$) and the UVM ($n = 8$) and 16 sessions from the RHM ($n = 9$), out of 18 possible sessions at each location across observers. (Because of technical problems in the eye data, we could not analyze the other sessions).

Microsaccade detection

A standard velocity-based detection algorithm (Engbert & Kliegl, 2003) was used to detect potential microsaccades in 2D velocity-space using a threshold of ≥ 6 SD above the average velocity per trial, for a minimum duration of 6 ms. Microsaccades were defined as saccades with an amplitude less than 1° of

visual angle. Eye positions were converted into degrees of visual angle, and microsaccades were characterized in terms of six parameters: onset time, offset time, peak velocity, horizontal amplitude (distance), vertical amplitude and angle (radians) to inform microsaccade direction.

Microsaccade rate analysis

Microsaccade counts were converted into hertz: microsaccades per second. Additionally, the resulting data were smoothed using a moving mean with a reasonable gaussian sliding window of 100 ms, which is within the 50 to 200 ms range usually implemented (Laubrock, Engbert, & Kliegl, 2005; Laubrock, Kliegl, Rolfs, & Engbert, 2005; Denison et al., 2019; Badde et al., 2020; Hung et al., 2022). Mean microsaccade rate throughout the trial for each condition was calculated per observer. Given the unequal sessions (15 sessions from the fovea and the UVM and 16 sessions from the RHM), a weighted average of microsaccade rates across observers was computed and used for all analyses. To analyze rate differences, we conducted a series of cluster permutation tests (Maris & Oostenveld, 2007) between attention and among location conditions. This nonparametric test compares time series data while correcting for multiple comparisons. We used a two-sided dependent samples cluster-based permutation test ($p < 0.05$) across observers. From each time point (and per observer) the time series data were compared and clusters of continuous significant t -values were identified ($p < 0.05$ or $a = 1.35$). We then summed the t -values to determine the size of each cluster. We shuffled the data (> 500 times) to obtain a normal distribution of cluster sizes to determine the significance of the clusters. From the cluster permutation tests we can ascertain time intervals of significant differences in the trial when comparing two conditions.

Microsaccade direction analysis

Microsaccade proportion was calculated for 16 polar angle bins (22.5° per bin), based on the angle (radian) parameter. We separately looked at microsaccades according to important trial segments: pre-target (precue onset to T1 onset), target (T1 onset to T2 offset), post-target (T2 offset to response-cue onset), and post-response (response-cue onset to 1000 ms after go-cue). We found that the microsaccade direction pattern was consistent across trial periods, thus for all following direction analyzes we collapsed across all trial segments. Microsaccade directionality was analyzed over the full trial sequence (precue onset to 500 ms after the response-cue) because the direction pattern

remained consistent throughout the trial with varying occurrence of microsaccades.

To investigate directional biases, we conducted a gaze analysis of the average gaze position for each trial segment and across the trial for each location (across attention conditions). To analyze the directionality of microsaccades in our dataset, we first conducted an omnibus test, which divides the circular distribution in half based on density differences. This test is specifically well suited for multimodal distribution of the data, similar to ours, and ultimately is a test to confirm the non-uniformity of our data. Microsaccade direction was then coarsely blocked in larger segments (67.5°) centered on the cardinal axes for analysis. Microsaccade occurrence in each segment was used (as in Badde et al., 2020) for comparison in a series of Holm-Bonferroni alpha-adjusted t -tests (Holm, 1979).

Results

Performance

In Fernández et al. (2019), the tilt thresholds revealed that observers ($n = 10$) showed the typical asymmetrical performance. Because we could recover microsaccade data for nine observers for two locations and eight observers for the third location, we confirmed equal variance among the three locations and performed a two-way repeated measures ANOVA with the participants who had data for all locations conditions ($n = 8$). Here (Figure 2A) we found the same main effect of location ($F(2,42) = 11.33$, $p = 0.0001$, $\eta^2 = 0.35$). Pairwise Holm-Bonferroni alpha adjusted t -tests ($\alpha = 0.05$) confirmed lowest thresholds for fovea (fovea vs. UVM $t(15) = 5.57$, $p = 0.00053$, compared to $a = 0.025$, $d = 1.64$), followed by RHM (RHM vs. fovea $t(15) = 6.33$, $p = 0.00013$, compared to $a = 0.0167$, $d = 1.75$), and highest for UVM (UVM vs. RHM $t(15) = 3.02$, $p = 0.0086$, compared to $a = 0.05$, $d = 0.38$). These findings are consistent with previous studies reporting contrast threshold at isoeccentric locations (e.g., Himmelberg et al., 2020). Regarding target discrimination (Figure 2B), we found no significant main effect of location ($F(2,42) = 2.4$, $p = 0.10$) or target ($F(1,42) = 0.02$, $p = 0.9$) and no significant interaction ($F(2,42) = 2.4$, $p = 0.10$). This result indicates that target discriminability was equivalent across the tested locations for these data. For reaction times there was neither a significant main effect of location ($F(2, 42) = 0.11$, $p = 0.9$) or target ($F(1, 42) = 0.46$, $p = 0.5$), nor an interaction ($F(2, 42) = 0.22$, $p = 0.8$).

Once discriminability was equated across locations, when compared to neutral trials, temporal attention improved discriminability (d') ($F(1,42) = 5.01$, $p = 0.03$, $\eta^2 = 0.09$), and there was no significant interaction

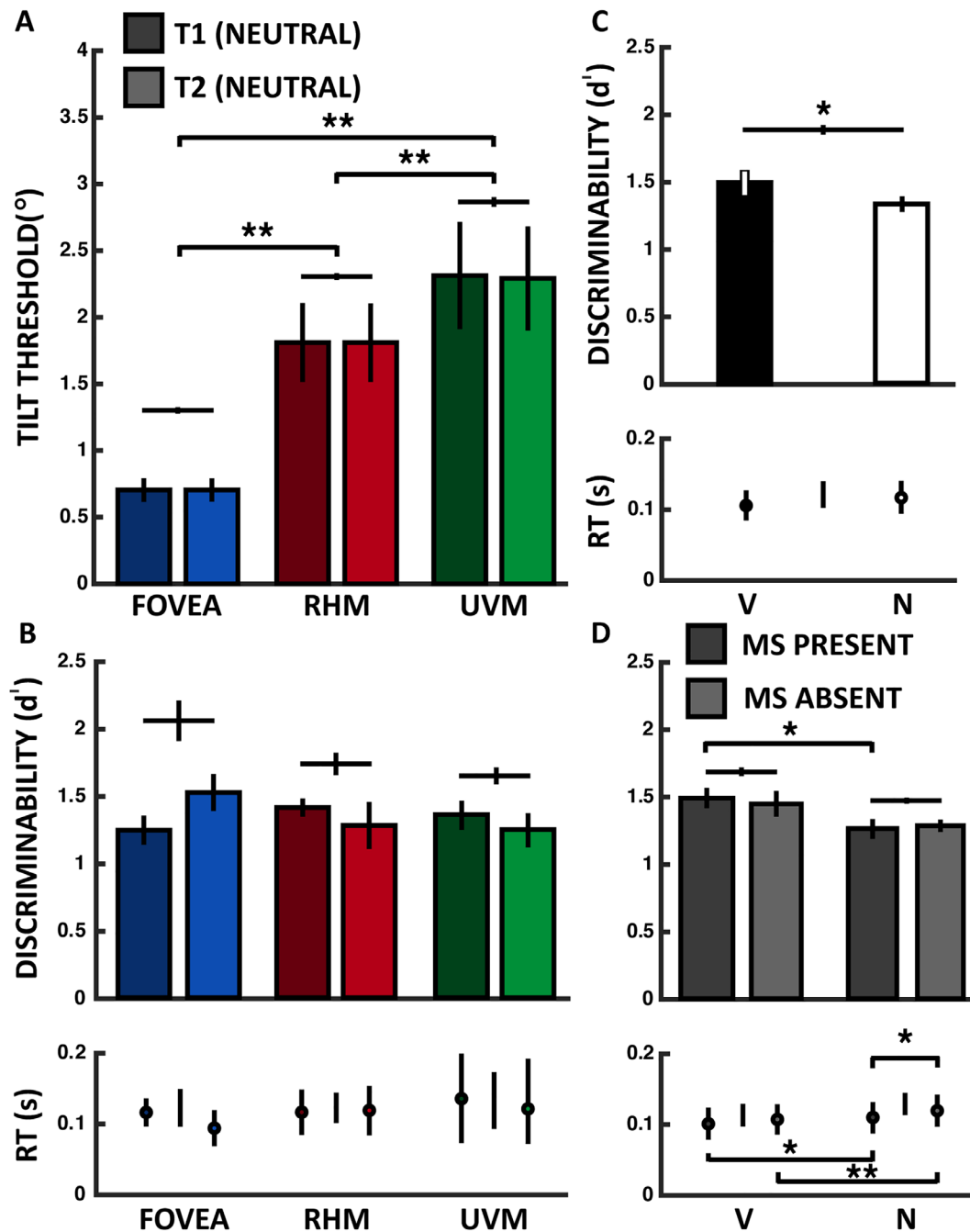


Figure 2. Temporal attention effects on behavior. (A) Mean orientation threshold for each target and location, neutral trials only ($n = 8$). Error bars between the bars represent the error of the differences between each respective condition. (B) Mean discriminability and reaction time for each target and location, neutral trials only. (C) Mean discriminability and response times across locations and targets. V = valid, N = neutral. (D) Mean discriminability and response times across locations and targets separated by whether or not there was a microsaccade present during the critical period (0–100 ms before target onset). Error bars in each panel represent the within-subject SEM with Cousineau correction. V = valid, N = neutral. * $p < 0.05$, ** $p < 0.01$.

between attention and location conditions ($F(2,42) = 0.15$, $p = 0.9$), and for reaction times there were neither main effect nor an interaction ($p > 0.1$). Thus there were no speed-accuracy tradeoffs. These results indicate that temporal attention benefits are comparable across critical cardinal locations and at fixation.

Microsaccade timing and behavior

Microsaccade suppression in anticipation of a target is a well-documented effect (Dankner et al., 2017; Denison et al., 2017; Amit et al., 2019; Abeles et al., 2020; Badde et al., 2020). During the microsaccade

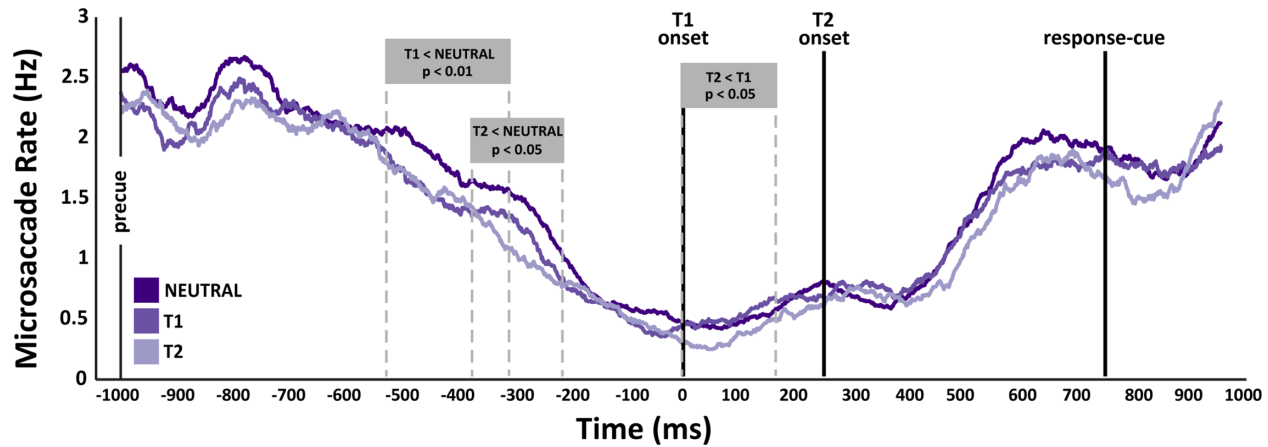


Figure 3. Microsaccade rate across isocentric locations. Group-average microsaccade rates. Microsaccade rates collapsed across isoecentric location conditions (RHM & UVM) as a function of trial time with respect to first stimulus onset. Gray dotted lines indicate boundaries of significance for time intervals from our cluster-permutation analysis. Microsaccade rates in neutral trials (dark purple), T1 trials (purple) and T2 (light purple).

pre-target inhibition time interval (-1000 to 0 ms, where 0 ms is target (T1) onset) there were fewer microsaccades overall as time approached the target display (~ 1.65 microsaccades). Trials were separated based on whether a microsaccade occurred during the critical period of 0 to 100 ms before target onset. Microsaccade occurrence during the critical period ($\sim 44\%$ of trials) did not significantly affect discriminability (d') in valid trials ($t(8) = 0.82$, $p = 0.44$) or neutral trials ($t(8) = 0.83$, $p = 0.43$) (Figure 2D). This was also the case for reaction time (ms) in valid ($t(8) = 2.15$, $p = 0.07$, $d = 0.09$), but there is a reaction time reduction in neutral trials ($t(8) = 2.90$, $p = 0.02$, $d = 0.12$) when microsaccades occurred. Performance (d') was higher for valid than neutral trials for trials with ($t(8) = 2.60$, $p = 0.03$, $d = 1.50$) or marginally without ($t(8) = 2.02$, $p = 0.08$, $d = 1.11$) microsaccades. Likewise, reaction times (ms) were faster for valid than neutral trials for trials with ($t(8) = 2.57$, $p = 0.03$, $d = 0.17$) or without ($t(8) = 3.63$, $p = 0.008$, $d = 0.20$) microsaccades. Thus the effects of temporal attention were the same regardless of microsaccade occurrence.

Microsaccade temporal patterns

Microsaccade rates were suppressed differentially depending on temporal attention manipulations. The precue refers to how we separate microsaccade rates, neutral is when participants received the neutral cue at the beginning of the trial, T1 is when participants were cued to T1 at the beginning of the trial, and T2 is when participants were cued to T2 at the beginning of the trial. (Specifically, microsaccades in T1 precued trials (Figure 3) were more suppressed than in neutral precued trials in anticipation of target presentation

(window: -538 ms to -314 ms, $p = 0.004$, $g = 0.28$) when collapsing nonfoveal locations (RHM and UVM, consistent with a previous study (Denison et al., 2019), which found that gaze stabilization can be a correlate of temporal attention deployment. As per the cluster analysis, additional significance intervals appeared while anticipating T2 (T2 vs. neutral, window: -391 ms to -225 ms, $p = 0.023$, $g = 0.59$) and between the target presentations (T1 vs. T2 window: -13 ms to 160 ms, $p = 0.019$, $g = 0.81$).

The overall microsaccade rate followed the typical dynamics across a trial (e.g., Rolfs, Kliegl, & Engbert, 2008; Yuval-Greenberg & Deouell, 2011; Denison et al., 2019) (Figures 4, 5). Microsaccade rate during the time of the precue averaged ~ 2.2 Hz. After this period the microsaccade rate steadily decreased between -500 ms and before the first target onset (0 ms), from a rate of ~ 1.8 Hz to a rate of ~ 0.3 Hz. This reduction is the “pre-target inhibition” (Dankner et al., 2017; Amit et al., 2019; Abeles et al., 2020; Badde et al., 2020), which indicates a strong effect of the stimulus timing expectation. A pronounced “post-target rebound” occurred in the 300 ms after the second final target presentation (Denison et al., 2019). After that steep rebound the microsaccade rate reached a steady rate of ~ 1.8 Hz.

In neutral trials (Figure 4A), microsaccade rates were significantly higher for RHM than foveal locations from the pre-target interval until after T1 presentation (window: -409 ms to 191 ms, $p = 0.031$, $g = 0.55$, except for a spurious 7 ms interval), in anticipation of the response-cue and after its offset (window: 398 ms to 957 ms, $p = 0.039$, $g = 1.13$). In attend T1 trials (Figure 4B), microsaccade rates were significantly higher for RHM than foveal locations during the stimulus onset asynchrony and briefly after the presentation of the

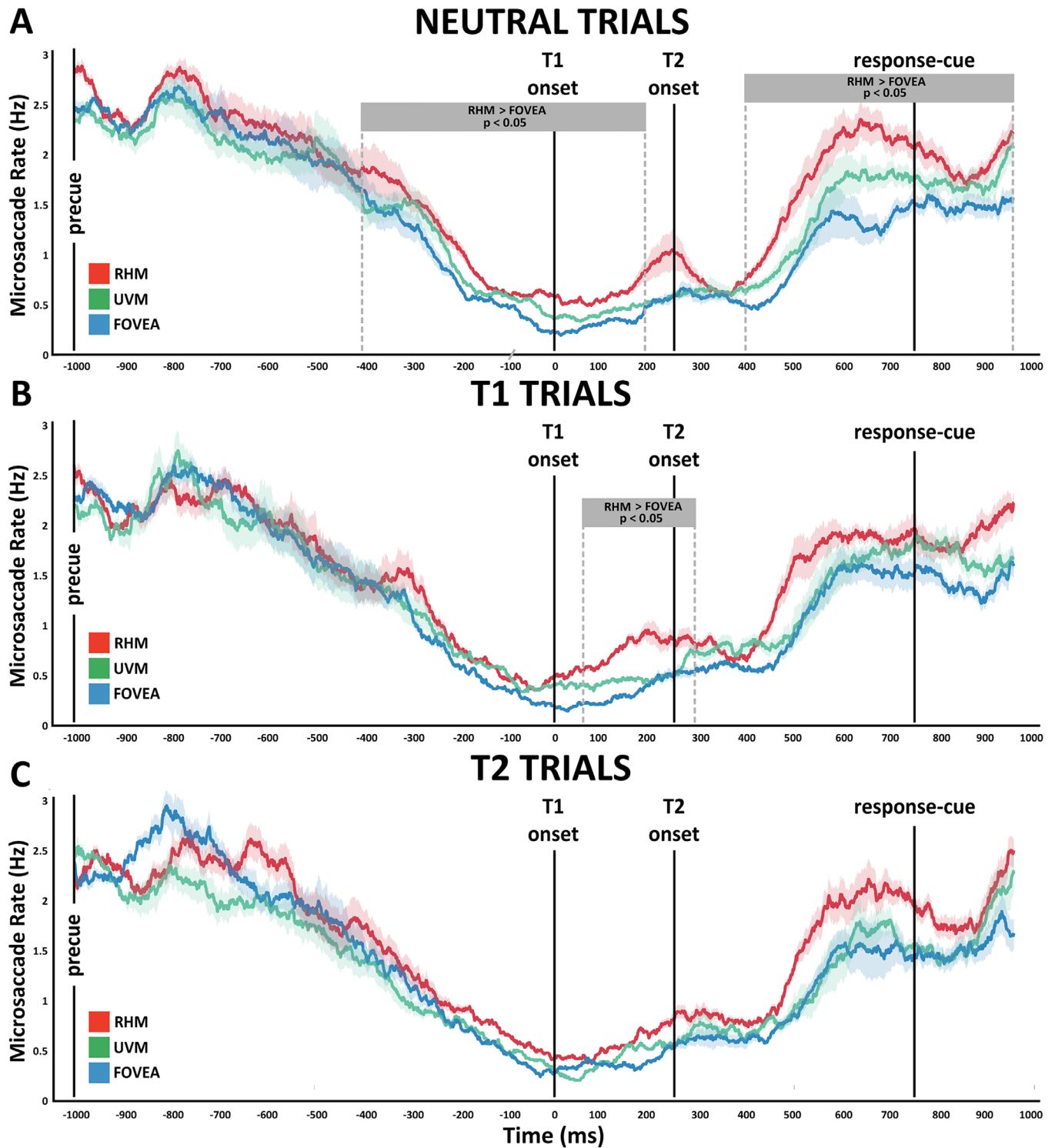


Figure 4. Microsaccade dynamics for each attention condition. (A) Microsaccade rates in RHM (red), UVM (green) and fovea (blue) location conditions in neutral trials. (B) Microsaccade rates in RHM (red), UVM (green) and fovea (blue) location conditions in T1 trials. (C) Microsaccade rates in RHM (red), UVM (green) and fovea (blue) location conditions in T2 trials. (A–C) Average microsaccade rates as a function of trial time with respect to first stimulus onset (T1 time zero). Corresponding color shading around the rate function represents the 68% confidence interval. Gray dotted lines indicate boundaries of significance for time intervals from our cluster-permutation analysis.

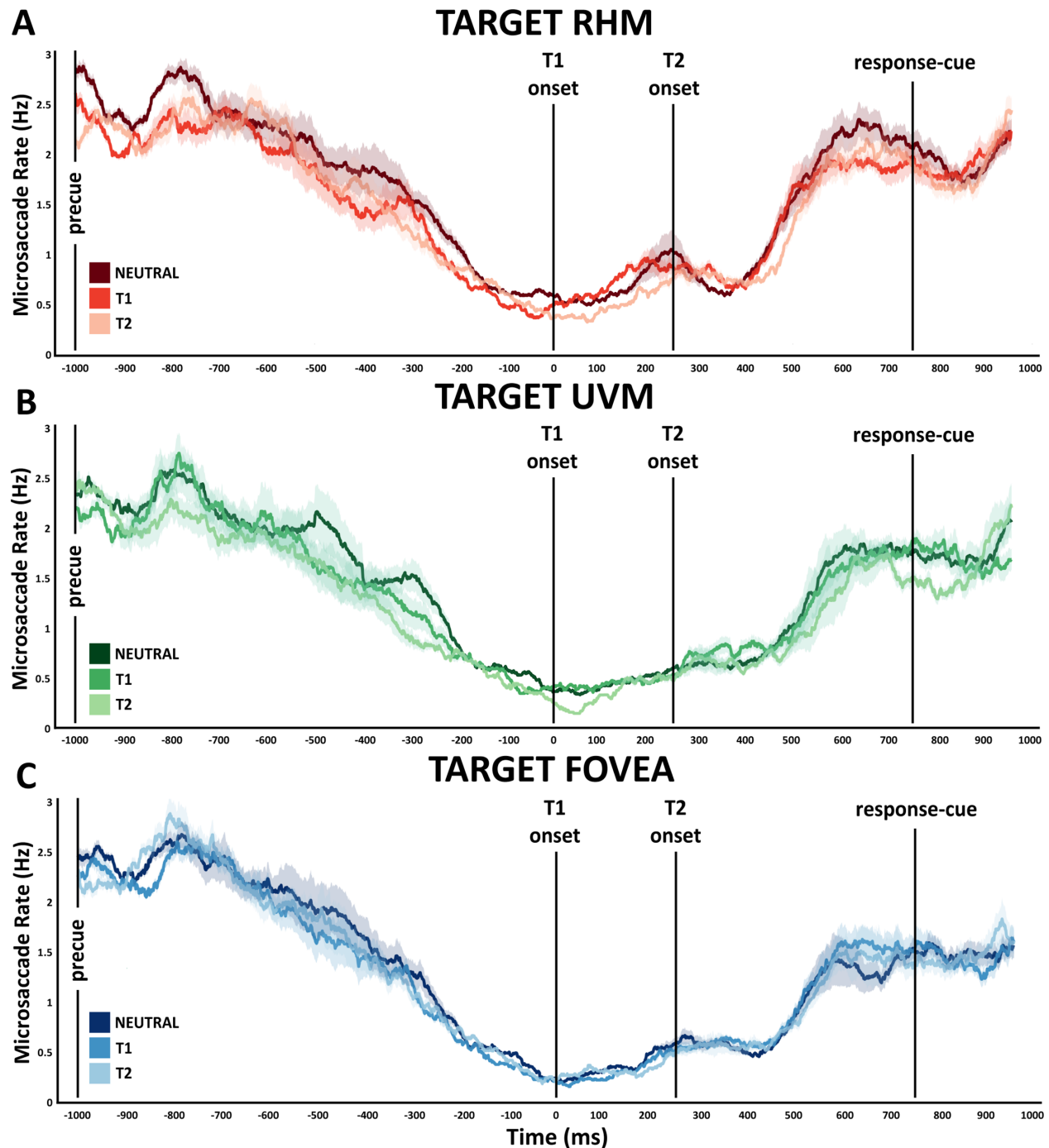


Figure 5. Microsaccade dynamics for each location condition. (A) Microsaccade rates in neutral (dark red), T1 (red) and T2 (light red) location conditions in RHM. (B) Microsaccade rates in neutral (dark green), T1 (green) and T2 (light green) location conditions in UVM. (C) Microsaccade rates in neutral (dark blue), T1 (blue) and T2 (light blue) location conditions in fovea. (A–C) Average microsaccade rates as a function of trial time with respect to first stimulus onset (T1 time zero). Corresponding color shading around the rate function represents the 68% confidence interval. Gray dotted lines indicate boundaries of significance for time intervals from our cluster-permutation analysis.

second target (window: 58 ms to 294 ms, $p = 0.046$, $g = 1.38$). In attend T2 trials (Figure 4C), there were no significant differences for microsaccade rates when comparing across locations. This result can be partially

due to longer microsaccade suppression in anticipation of T2 (see Figure 4C: 0–200 ms). In summary, microsaccade rate varied across target locations, it was most frequent in the RHM and least in fovea.

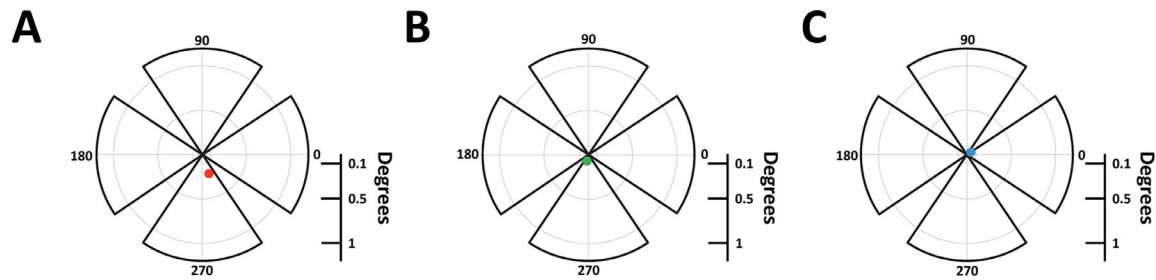


Figure 6. Gaze position across the trial sequence. (A) The average gaze position for RHM blocks (0.223°) is depicted by a red dot near the center. (B) The average gaze position for UVM blocks (0.07°) is depicted by a green dot near the center. (C) The average gaze position for fovea blocks (0.052°) is depicted by a blue dot near the center.

Plotting the same data to compare microsaccade rates within a location illustrates that rates are similar across attention conditions for each of the three locations (Figures 5A–C). For RHM and UVM overall there were less microsaccades for attend T2 than attend T1 and neutral trials, but no significant differences emerged. These results indicate the generalizability of the overall temporal profile of microsaccades at each location.

Microsaccade direction patterns

We performed a gaze analysis by averaging gaze angle position for each location for relevant trial segments and across the entire trial. We plot the mean gaze for each location for the whole trial, as the mean (as well as the median) of the gaze was consistent throughout the trial (Figure 6). For the fovea and the UVM, the gaze is at the center, and for the RHM, it is slightly below (0.2°) the center.

For microsaccade direction, which was based on radian angle from the current eye position, there was a higher number of microsaccades towards the upper hemifield. All omnibus tests (Hodges-Ajne test: Zar, 1999) confirmed that these microsaccade direction distributions were nonuniform around a circle ($p < 0.001$). After the uniformity test, microsaccade direction was blocked in segments (67.5°) centered on the cardinal axes (Figure 7). Microsaccade count in each of these four segments collapsed across attention conditions and target location was used for comparison. Pairwise holm-Bonferroni alpha adjusted t -tests revealed that microsaccades were significantly or marginally higher at left horizontal meridian (LHM), RHM, and the UVM than at lower vertical meridian (LVM) (Figure 7: LHM versus LVM $t(14) = 2.85$, $p = 0.0128$, compared to $\alpha = 0.0167$, $d = 1.43$; RHM vs. LVM $T(14) = 2.67$, $p = 0.0182$, compared to $\alpha = 0.025$, $d = 1.34$; UVM vs. LVM $t(14) = 1.77$, $p = 0.0993$, compared to $\alpha = 0.05$, $d = 0.88$). Given that gaze position was quite precise close to fixation (within $\sim 0.2^\circ$), throughout the trial

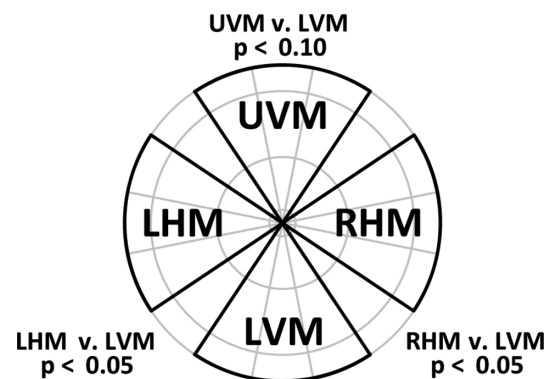


Figure 7. Direction segmentation. Microsaccade direction divided into four segments (LHM, UVM, RHM, & LVM). This segmentation results in 67.5° wedges.

(as well as when analyzed for separate intervals of the trial), it is unlikely that corrective microsaccades for fixation are responsible for the upper hemifield bias of microsaccade direction.

Additionally, there are no differences in microsaccade direction dependent on target location (Figures 8A–C), and direction patterns are consistent regardless of attention manipulations (Figures 8C–F). Hence, microsaccades were directed toward the upper hemifield throughout the trial and across conditions.

Discussion

We investigated microsaccades rate and direction as participants completed a temporal attention task. As stimulus location was blocked, there was no location uncertainty regarding the target location. Critically, behavioral temporal attention effects were indistinguishable with and without microsaccades (Figure 2), indicating that microsaccades are not mediating the effect of temporal attention. However, we found that microsaccade rates were modulated by the

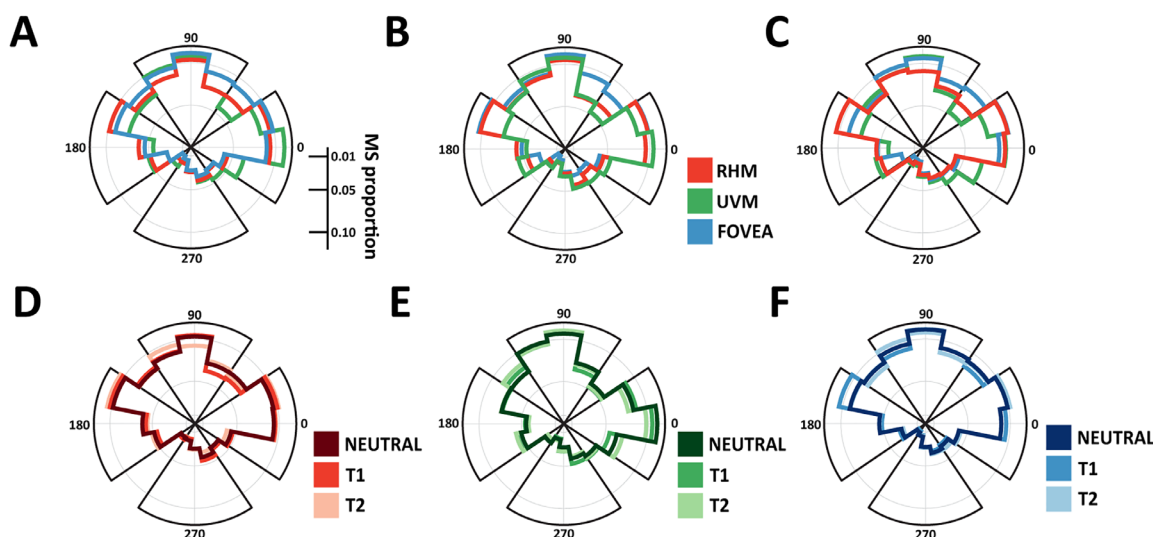


Figure 8. Microsaccade direction. Microsaccade direction divided into 16 slices, with the slices centered on the cardinal meridians. Slices indicate microsaccade proportion for the overall trial sequence. (A) Microsaccade direction for neutral trials, red line for RHM, green line for UVM, and blue line for fovea. (B) Microsaccade direction for T1 trials. (C) Microsaccade direction for T2 trials. (D.) Microsaccade direction for RHM target location, dark red line for neutral, medium red for T1, and light red for T2. (E) Microsaccade direction for UVM target location, dark green line for neutral, medium green for T1 and light green for T2. (F) Microsaccade direction for fovea target location, dark blue line for neutral, medium blue for T1, and light blue for T2.

location for the neutral (Figure 4A) and when attending the first interval (T1) (Figure 4B). In particular, microsaccades were most frequent when targets appeared at the RHM location and most infrequent when they appeared at the fovea (Figure 4). In anticipation of the target display, during the pre-target inhibition period, microsaccade rate decreased, but temporal attention (when selectively attending to T1 or T2 trials) caused microsaccade rates to decrease earlier than in the neutral trials (Figure 3). Additionally, we observed an overall tendency of microsaccade direction towards the upper hemifield (Figure 7), which is inconsistent with the commonly reported horizontal bias (Engbert, 2003; Yuval-Greenberg et al., 2014). Neither target location nor temporal attention manipulations modulated this direction effect.

Performance fields factors

Discriminability is better along the horizontal than the vertical meridian and along the lower than the upper vertical meridian in many visual tasks (e.g., Carrasco et al., 2001; Cameron, Tai, & Carrasco, 2002; Abrams, Nizam, & Carrasco, 2012; Baldwin et al., 2012; Himmelberg et al., 2020). Optical and retinal factors only account for a small portion of polar angle behavioral asymmetries (Kupers, Carrasco, & Winawer, 2019; Kupers, Benson, Carrasco, & Winawer, 2022). Cortical differences (Liu, Heeger, &

Carrasco, 2006; O'Connell et al., 2016; Silva et al., 2018; Benson, Kupers, Barbot, Carrasco, & Winawer, 2021; Himmelberg et al., 2021; Himmelberg, Winawer, & Carrasco, 2022) explain a larger variance of these performance differences (Benson et al., 2021; Kupers et al., 2022). Moreover, individual differences in overall V1 size and differences in cortical magnification at cardinal locations correlate with contrast sensitivity (Himmelberg et al., 2022). Performance differences across polar angles, as well as their retinal and cortical correlates, are well characterized (review, Himmelberg, Winawer, & Carrasco, 2023). The current study adds microsaccades as an oculomotor factor to be investigated with regard to polar angle asymmetries. We found that the microsaccade rate varies with location and attention conditions but that their direction does not; there was an overall tendency to microsaccade toward the upper visual field.

Microsaccade rates vary with location and attention conditions

Location modulated temporal patterns for microsaccade rates for neutral (attend to both intervals) and T1 (attend to the first interval) (Figures 4A-B). More specifically, in neutral trials there were fewer microsaccades when the target was in the fovea than in the RHM during pre-target inhibition, partially through the target presentation and before and

after the response-cue, regardless of the attention condition (Figure 4A). The UVM microsaccade rate was intermediate between the fovea and RHM microsaccade rates during most of the trial sequence (Figure 4). Given that the fovea is the location of highest acuity, performing microsaccades during a brief stimulus presentation could hinder performance especially there by distorting the retinal image. Hence, overall, microsaccade rates in the fovea are the lowest (Figure 4) and the least variable to temporal attention effects (Figure 5).

In all location and attention conditions, we observed pre-target oculomotor inhibition, which serves a functional role in perceptual performance in visual tasks (i.e., avoiding [micro]saccades and blinks) while anticipating a predictable target may enhance subsequent target perception. Saccades and blinks are known to cause a temporary loss of visual input because of physical occlusion, image blur, or masking (Zuber & Stark, 1966; Martinez-Conde, Otero-Millan, & Macknik, 2013) and also be accompanied by active suppression in sensory cortices (blink suppression (Volkman, Riggs, & Moore, 1980) and saccadic suppression (Volkman, Schick, & Riggs, 1968; Abeles et al., 2020)).

Humans inhibit microsaccades in anticipation of a predictable brief stimulus (e.g., Dankner et al., 2017; Denison et al., 2019; Amit et al., 2019; Abeles et al., 2020; Badde et al., 2020). The relation between high target predictability, greater microsaccade suppression, and higher task performance is well documented (Dankner et al., 2017; Amit et al., 2019; Badde et al., 2021). For example, neurotypical adults inhibit microsaccades more before a predictable target than a random target regardless of target modality; for visual (Dankner et al., 2017; Amit et al., 2019), auditory (Abeles et al., 2020), and tactile (Badde et al., 2020) stimuli. For neurotypical adult observers the higher the suppression during, shortly before, or after a visual (Dankner et al., 2017; Amit et al., 2019) or tactile (Badde et al., 2020) target, the better the performance in detection and discrimination tasks. In contrast, adults with attention disorders (ADHD) do not differ in their microsaccade suppression between predictable and unpredictable visual stimuli. This lack of inhibition before predictable stimuli could be related to poorer performance in people with ADHD in some visual tasks (Fried et al., 2014; Dankner et al., 2017).

Temporal attention benefits go beyond the benefit of temporal expectation. Deploying temporal attention increases the stabilization of the eye (less microsaccades) at the time of the cued target interval, compared to neutral trials, for which there is only an expectation effect (Denison et al., 2019). The current data replicate this finding when collapsing across stimulus locations (Figure 3). Furthermore, here,

during the pre-target inhibition, the suppression for T1 occurred earlier than the suppression for T2. This finding aligns with the fact that the microsaccade rate is suppressed according to the length of the foreperiod; the faster a stimulus appears the earlier the suppression happens (Amit et al., 2019; Badde et al., 2020). Even though microsaccades were more inhibited before target presentation with attention than in the neutral condition, we note that behavioral effects for the neutral and attention conditions were indistinguishable with and without microsaccades (Figure 2). This result shows that microsaccades accompany but do not drive visual effects.

Microsaccades directed to the upper hemifield regardless of target location or attention condition

Our task design allowed us to observe microsaccade temporal patterns and directional patterns as a function of attention manipulation and performance field location with no spatial uncertainty. We observed an upper hemifield bias irrespective of target location. These results differ from the commonly reported horizontal bias for microsaccade direction. In most studies reporting horizontal bias the stimuli were presented simultaneously and only placed along the horizontal meridian (e.g., Engbert & Kliegl, 2003; Laubrock et al., 2005; Engbert, 2006; Liang et al., 2005; Moshel et al., 2008; Yuval-Greenberg et al., 2014; Raveendran, Krishnan, & Thompson, 2020), but only few report the boundary (wedge in degrees) for their analyses (Engbert et al., 2003: 30°; Yuval-Greenberg et al., 2014: $\pm 22.5^\circ$; Raveendran et al., 2020: 90°). Thus the previously documented horizontal bias could be due to previous task designs with stimuli regularly presented on the left and right of fixation. Moreover, some have observed horizontal microsaccades directed toward the cued location immediately after an attention cue but then away from the cued location 300 ms after the cue presentation (Rolfs, Engbert, & Kliegl, 2004; Laubrock et al., 2005). This pulling mechanism toward and away from the cued location has only been reported along the horizontal meridian. In our study, the absence of microsaccades in the downward direction suggests that the push-pull mechanism only applies along the horizontal meridian.

Instead of a horizontal bias, we observed an upper hemifield bias. In the current study, we presented stimuli at the fovea as well as along the RHM and the UVM and only at one location per block. Does this stimulus configuration explain the preponderance of microsaccade towards the upper hemifield? We think this is unlikely as we see the same upper hemifield bias even when the stimuli were presented at the

fovea or the RHM. Moreover, had this been the case, we should have observed few microsaccades to the left, where the stimulus was not presented. But we do see microsaccades to the left and right at similar levels. Some have observed vertically directed microsaccades for small targets at fixation, but always less frequent than in the horizontal direction (Liang et al., 2005; Moshel et al., 2008). Interestingly, even when participants have block-level knowledge of target location, microsaccade direction was not pulled towards that target location. By showing that gaze was precise (within $\sim 0.2^\circ$) near the fixation, we ruled out that gaze position could be responsible for the observed upper hemisphere bias. Had the eye drifted downward (for example, because of fatigue), microsaccades could have been directed upward to correct gaze back to fixation.

The location at which performance is usually the worst (UVM) is included in the area showing highest frequency of microsaccades directional bias. Thus our results support our hypothesis that microsaccade direction is influenced by typical behavioral performance asymmetries. The system could have a meta-knowledge of these performance asymmetries and try to sample the worse location (UVM) more frequently by using microsaccades. This finding is reminiscent of the finding that in visual search tasks humans take into account how the sensitivity of their visual system varies across the visual field; when looking for a small target embedded in noise, the frequency of the first saccades is higher towards the UVM where discriminability is poor (Najemnik & Geisler, 2008).

The upper hemifield bias has also been observed in a recent study when targets are presented simultaneously at the four cardinal locations (Purokyastra, Roberts, & Carrasco, 2020). In an orientation discrimination task with constant stimulus resulting in heterogeneous discriminability or with stimulus parameters adjusted to yield homogeneous discriminability across all four cardinal locations, the upper hemifield bias for microsaccades emerged in both cases. This bias was observed during the pre-stimulus interval, although there is uncertainty regarding the location at which the target and three distractors will appear, and after the stimulus display, when there is no longer target location uncertainty. Together, these studies show that the upper hemifield bias in microsaccades is not due to location uncertainty and that when stimuli are presented along the vertical meridian, this upper hemifield bias occurs.

Notwithstanding the fact that in the present study we equated discriminability around the visual field, this upper visual field bias could be related to the discrimination asymmetries in the visual field or the corresponding asymmetries in visual cortex. The pronounced difference in microsaccades directed towards the lower and upper visual hemifields could be an oculomotor attempt to help compensate for the

often observed horizontal-vertical and vertical meridian asymmetries in performance (review, Himmelberg et al., 2023). That is, the visual system could have a meta-knowledge of these typical performance asymmetries and try to sample the UVM more frequently by using microsaccades, during times at which microsaccades would not impair perception. For example, in some studies, it has been found that microsaccades are directed to the location where the target had just appeared. These changes likely reflect functional dynamics of the oculomotor system during maintenance and readout of information (Hung & Carrasco, 2022; Hung, Barbot, & Carrasco, 2022; van Ede et al., 2019).

Conclusion

Temporal attention improves performance similarly regardless of target location and regardless of whether trials contain microsaccades. First, we replicated the result that microsaccades are suppressed earlier in attention trials (attend T1 and attend T2) during the pre-target interval and target presentation than neutral trials (Denison et al., 2019). Second, we show that microsaccade rate is dependent on target location, with more microsaccades occurring when the target appeared at the horizontal peripheral location (RHM) than at the fovea, and this effect is most pronounced in the neutral trials. Third, microsaccade direction had an overall upper hemifield bias, regardless of target location or attention condition. This pattern differs from the commonly reported horizontal bias. The present findings could have translational applications in human factors and clinical populations. For instance, they could inform possible diagnostic measures for oculomotor abnormalities in special populations, including people with amblyopia or attentional disorders (ADHD, ADD), which could affect visual performance.

Mapping performance at different locations of the visual field is important for understanding how visual resolution and the oculomotor system interact with other important visual mechanisms, such as attention. Indeed, there is a growing interest in the study of the role of saccadic precision and latency (Greenwood, Szinte, Sayim, & Cavanagh, 2017), as well as the effects of presaccadic attention (Hanning, Himmelberg, & Carrasco, 2022; Kwak, Hanning, & Carrasco, 2022) around the visual field. To understand naturalistic vision and behavior, we must investigate how these effects work in tandem to give rise to our everyday visual experiences.

Keywords: temporal attention, microsaccades, performance fields

Acknowledgments

The authors thank Aysun Duyar and other Carrasco lab members for helpful feedback and comments on the manuscript.

Supported by the National Eye Institute of the National Institutes of Health under Award Number R01EY019693, R01-EY027401 (to MC), Diversity Supplement 3R01EY019693-08S (to HP), and NINDS F99NS120705 to AF.

Commercial relationships: none.

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